Reliable Estimation of Automotive States Based on Optimized Neural Networks and Moving Horizon Estimator

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***Abstract*—Accurate estimation of vehicle sideslip angle and attitude angles are essential for the safety control and lateral behaviour of driving performance. In this article, the variation of wheels cornering stiffness is considered for sideslip estimation and addressed by introducing a recursive least squares approach. Based on the nonlinear vehicle dynamic model and the investigated coupling effect between lateral and longitudinal velocity, an optimized moving horizon estimator is proposed to obtain the vehicle sideslip angle, in which an iteration decent algorithm is integrated. Furthermore, a framework, consisting of inertial navigation system measurements, a dual neural network and a square-root cubature Kalman filter, is designed, such that the influence of sensor noise and varied maneuvers are alleviated when estimating the system states. Finally, extensive simulation and field experiments are carried out on different driving scenarios to verify the effectiveness of the developed method. The obtained results clearly indicate the satisfactory estimation accuracy of the designed strategy, superior to the existing estimation methods, such as sole neural networks methods and Kalman-based filters.**

***Index Terms*—Automotive system, cornering stiffness, moving horizon estimator (MHE), neural network, sideslip angle, state estimation.**

I. Introduction

ACTIVE safety systems, such as anti-lock brake system and electronic stability control (ESC), have been developed in modern automobiles and can effectively prevent a large portion of road accidents [1]. However, the system performance is highly dependent on the accurate knowledge of vehicle states, such as the lateral vehicle speed and sideslip angle, for which, direct measurement is unfortunately difficult due to either technical or economic reasons. Hence, different approaches have been widely considered to estimate vehicle states [2], [3], [4]. However, the estimation accuracy still needs to be further improved, especially for vehicles operated under a wide operation range.

The sideslip angle directly affects the vehicle dynamics, whose estimation has attracted considerable research efforts [5]. For the current approaches, dynamic models with different assumptions are usually established and corresponding observers are then designed accordingly, which are unfortunately dependent on the accurate information of vehicle parameters and tire-road conditions. To adapt to various road conditions during actual moving, the states of tire–road friction, road bank angle, and vehicle roll angle are also investigated [6]. A Takagi–Sugeno (T-S) fuzzy observer is designed to estimate both the steering angle and the sideslip angle, which demonstrates good robustness against parametric uncertainties and unknown inputs by solving linear matrix inequalities [7]. Besides, some nonlinear tire models, such as Dugoff and Pacejka's magic formula [8], are employed to approximately describe the lateral force within the system.

Other methods based on the kinematic model calculate the states of the system by fusing sensor information, such as the measurements for the yaw rate, lateral acceleration, and so on. Unfortunately, these methods are often vulnerable to sensors bias and measurement errors caused by severely driving conditions. Especially, the velocity estimated by additional sensors, such as global positioning system (GPS), is difficult to obtain when the studied vehicle is braking on low friction surfaces [9], [10]. Besides, a hybrid approach, consisting of recurrent neural networks (RNNs) and a kinematic vehicle model, was presented in [11] to estimate the sideslip angle with the aid of a physical model under different situations. Boada et al. [12] adopted the pseudoside slip angle obtained from one-layer fully connected neural network and Kalman filter (KF) to predict the sideslip angle. Though the aforementioned approaches are effective in some situations, they face the same difficulty of applying to practical vehicles with more complex dynamics, due to the limitation of the network architecture.

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Due to the reason that the lateral force and slip properties are considered in dynamic model-based techniques, these methods are usually more appropriate for high-speed and high-slip driving conditions. On the contrary, the kinematic model-based methods perform better when the slip angle is sufficiently small under low speed maneuvers [13]. Based on these facts, new approaches based on the combination of the aforementioned methods are also developed to bring together their advantages [14], so that the proposed approaches achieve satisfactory performance for various scenarios. The longitudinal dynamic model includes many uncertainties, such as the road slope and other forces [15], meanwhile, the variation of driving speed also largely affects the lateral dynamics. An accurate vehicle velocity estimation strategy was proposed in [1] for the braking process, whose gains change accordingly so as to accommodate different high-slip cases. So far, few papers focus on the study of the relationship and the coupling effect between the lateral and the longitudinal velocity during different maneuvers. In addition, uncertainties within the system, especially those associated with the variation of the sideslip angle and the longitudinal dynamics, need to be carefully considered when designing an estimator for the system states.

The cornering stiffness varies with the road surface condition and tire–road friction coefficient, for which an estimator was designed in [16] by exploiting the understeering characteristics of vehicles. To adapt to different road conditions, some recursive-based algorithms are proposed to estimate the cornering stiffness efficiently [17]. After some careful investigation, it is believed that the estimation for the vehicle sideslip angle, especially under the scenarios with various road surfaces and different tire–road friction coefficient, plays an essential role to implement efficient and safe driving, and needs to be further studied.

For the aforementioned estimation problem, the currently available techniques can be summarized as KF-based methods [18], nonlinear observers [19], parameters optimization strategies [20], fusion estimation approaches [21], and so on. Specifically, based on modeling techniques, different filters are introduced to construct estimators for vehicles states, whose performance highly relies on the accuracy of the established model. Although the extended Kalman filter (EKF) and unscented Kalman filter (UKF) have shown promising results, the accuracy of Kalman-based filters is limited directly by the prior knowledge of the process and measurement noise. In order to guarantee satisfactory performance for state estimation, especially in the presence of various uncertainties and disturbance, such as accumulate error in inertial navigation system (INS), a finite-memory estimator is proposed with recent finite measurements [22].

On the other hand, different from one-step recursion in EKF and UKF, a window covering the most recent measurements is utilized to optimize state estimation in moving horizon estimator (MHE). MHE has been employed to obtain reliable estimation for system states and parameters, and to some extent attenuates the effect of system unmodeled disturbance [23], [24], [25]. For nonlinear systems with unknown noise statistics, an expectation–maximization based MHE has been introduced [26], [27], for which, a suitable optimization algorithm is of kernel importance to solve the constrained problem. Different descent algorithms, based on gradient, conjugate gradient, and Newton methods, respectively, have been proposed in [28]. It is proved that the computational cost can be reduced by implementing a feed-forward neural network [29], and the conditions are established to ensure the stability of the estimation technique. Nevertheless, how to make the tradeoff between computation effort and performance indices is still a challenging and open problem. It has been verified that a prediction value can be provided by MHE to bridge the gap, especially for a multirate system with no measurement [30], [31], [32].

More recently, some optimization strategies are designed based on maximum likelihood analysis [33] and learning-based methods [34]. It has been proved that a long short-term memory (LSTM) neural network presents the ability of overcoming vanishing and exploding problems existed in traditional RNN, and a real-time multivariate LSTM-RNN model was constructed in [35]. Due to this reason, the LSTM is used to establish the relationship between KF gains and the observations. In addition, the LSTM has a capability of capturing the dependence among vehicle states and integrating with nonlinear filters to deal with the uncertainty in the estimation [36].

Motivated to provide satisfactory estimation for automotive states, and inspired by the previous methods, this article investigates the application of different iteration decent algorithms, and then designs a reliable estimation strategy for automotive sideslip angle and other states, which consists of optimized neural networks and MHE, for which, some practical rules are established to alleviate the influence caused by the sideslip and INS noise. Compared with other results accounting for various driving conditions, the designed algorithm yields more efficient and effective estimation for vehicle states. Specifically, the main contribution of this article lies in the design and improvement of a nonlinear estimator for vehicle states, specifically in the following aspects, which distinguish it from currently available results.

1) A multilayer estimation framework is proposed by integrating the MHE algorithm into vehicle dynamic model layer, with an optimized neural network in sensors layer and a square-root cubature Kalman filter (SCKF) in the layer of parameters optimization. In addition, the variation of cornering stiffness and different driving maneuvers are also carefully considered when designing the estimator.
2) With regard to the different characteristics of lateral and longitudinal motion, the LSTM neural networks (a dual-LSTM) are developed separately to establish the relationship and the networks structure are designed according to the different system models. Therefore, our strategies enable to learn the error properties of lateral velocity and sideslip angle with less data and lower cost GPS measurement, so as to guarantee the estimation accuracy for the vehicle states.
3) The proposed novel estimation method is thoroughly tested by both simulation and experiments under various driving conditions similar to practical scenarios. The advantage of the proposed method is clearly verified through

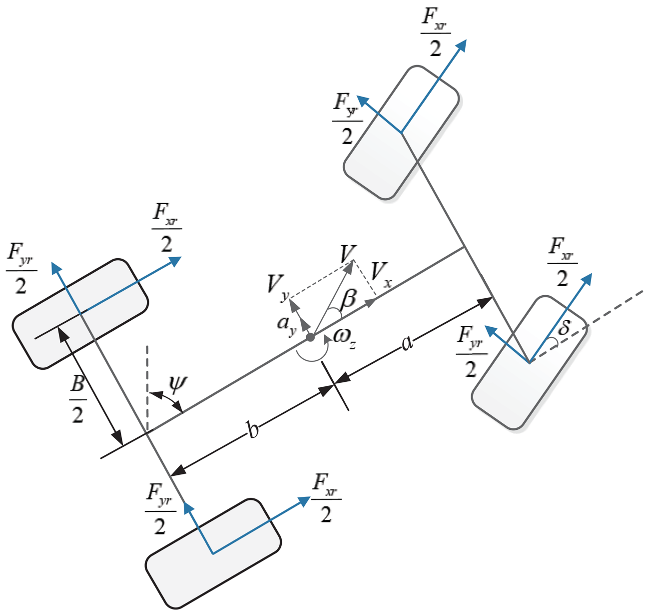

Fig. 1. Vehicle dynamic model.

some comparative experimental data with current Elman neural network and Kalman-based filters.

The rest of this article is organized as follows. In Section II, the vehicle dynamic model and the developed strategy are described in detail. Then, for the vehicle states, a nonlinear estimator is proposed in Section III. In Section IV, both simulation and experimental results are presented and analyzed to show the superior performance of the designed estimation method. Finally, Section V concludes the article.

II. Vehicle Model and Problem Formulation

To facilitate the subsequent estimation algorithm design for the vehicle states, the system dynamic model, including the vehicle motion dynamics, the tire model, and so on, is introduced and analyzed in detail. It is demonstrated that the wheel angular velocity measurements and traction or braking torque can be used to estimate the vehicle longitudinal velocity. Yet, the assumption of no longitudinal or lateral slips is needed, which is actually not very practicable for real applications, especially when the vehicle is sliding or locking on some low friction road surfaces. In general, the model-based methods are superior in computational efficiency but vulnerable to parameters variations [37].

A. Vehicle Dynamic Model

The illustration for the vehicle dynamic model is depicted in Fig. 1, where the average tire forces in the front and rear axle are considered. The related coordinate systems are defined as East–North–UP (n frame) and vehicle body frame (b frame). The dynamic equations of vehicle longitudinal, lateral, and yaw motion are as follows:

$$m\dot{v}_x = F_{xf}\cos\delta - F_{yf}\sin\delta + F_{xr} + m\omega_z \cdot v_y \tag{1}$$

$$m\dot{v}_y = F_{xf}\sin\delta + F_{yf}\cos\delta + F_{yr} - m\omega_z \cdot v_x - mg\sin\phi \tag{2}$$

$$I_z \cdot \dot{\omega}_z = a[F_{xf}\sin\delta + F_{yf}\cos\delta] - bF_{yr} \tag{3}$$

where v_x is the longitudinal velocity at the center of gravity (COG) and v_y is the lateral velocity. ω_z represents the yaw rate around COG, with the roll angle ϕ defined accordingly. m is the total mass and I_z is the inertial yaw moment. a and b are the distance from COG to the front and the rear axle, respectively. δ denotes the steering angle. B is the track width.

B. Tire Model and Cornering Stiffness Estimation

To describe the relationship among tire's lateral force, tire sideslip angle and road friction, different tire models have been set up in the previous research [37]. The Dugoff tire model is employed to represent the nonlinear forces related to different maneuvers. Assuming pure lateral slip, the tire lateral force is given by

$$F_{yi} = -C_i \tan\alpha_i f(\lambda_i) \tag{4}$$

$$f(\lambda_\mathrm{i}) = \begin{cases} \lambda_i(2-\lambda_i), & \text{if } \lambda_\mathrm{i} < 1 \\ 1, & \text{if } \lambda_\mathrm{i} \geq 1 \end{cases} \tag{5}$$

$$\lambda_i = \frac{\mu F_{zi}[1-|\tan\alpha_i|]}{2C_i\,|\tan\alpha_i|} \tag{6}$$

where C_f and C_r are the front and rear tire cornering stiffness, with the subscripts $i = f, r$. The parameter μ is the friction coefficient and it is related to the cornering stiffness. The normal force F_{zj} for the jth wheel can be presented as follows:

$$\begin{aligned}
F_{z1} &= \frac{m(gb - h_{CG}a_x)}{2(a+b)} - \frac{mh_{CG}a_y(gb - h_{CG}a_x)}{(a+b)Bg} \\
F_{z2} &= \frac{m(gb - h_{CG}a_x)}{2(a+b)} + \frac{mh_{CG}a_y(gb - h_{CG}a_x)}{(a+b)Bg} \\
F_{z3} &= \frac{m(ga + h_{CG}a_x)}{2(a+b)} - \frac{mh_{CG}a_y(ga + h_{CG}a_x)}{(a+b)Bg} \\
F_{z4} &= \frac{m(gl_i + h_{CG}a_x)}{2(a+b)} + \frac{mh_{CG}a_y(gb + h_{CG}a_x)}{(a+b)Bg}
\end{aligned} \tag{7}$$

where j is 1, 2, 3, and 4, and represents the front left, front right, rear left, and rear right wheels, respectively. h_{CG} denotes the height of COG, and a_x is the longitudinal acceleration. Recursive least squares (RLS) algorithm is adopted to identify the correct description of the vehicle on various roads with different conditions, especially when the lateral acceleration reaches a certain value. The front and the rear tire slip angles, denoted as α_f and α_r, can be calculated as follows:

$$\alpha_f = (v_y + a\omega_z)/v_x - \delta, \alpha_r = (v_y - b\omega_z)/v_x. \tag{8}$$

When the tire sideslip angle and the steering angle are sufficiently small, by integrating (2), (4)–(6), the lateral fores at the front and rear are represented as follows:

$$F_{yf} = -C_f M_f(\alpha_f)\alpha_f, F_{yr} = -C_r M_r(\alpha_r)\alpha_r \tag{9}$$

where $M_i(\alpha_i) = \frac{\tan\alpha_i f(\lambda_i)}{\alpha_i}$. When the $M_i(\alpha_i)$ is replaced by M_i for simplicity, (2) can be further expressed as follows:

$$\begin{aligned} ma_y &= m\dot{v}_y + \omega_z \cdot v_x \\ &= C_f M_f \left(\delta_f - \beta - \frac{a\omega_z}{v_x}\right) \\ &\quad - C_r M_r \left(\beta - \frac{a\omega_z}{v_x}\right) - mg\sin\phi. \end{aligned} \tag{10}$$

When $C_f M_f + C_r M_r = \Xi$, the sideslip angle is calculated as follows:

$$\beta = \frac{1}{\Xi} \left[C_f M_f \left(\delta_f - \frac{a\omega_z}{v_x}\right) + C_r M_r \frac{b\omega_z}{v_x} - mg\sin\phi - ma_y\right] \tag{11}$$

where a_y represents the lateral acceleration.

Based on the above discussion, (3) can be written as follows:

$$\begin{aligned} I_z \cdot \dot{\omega}_z = aF_{yf} - bF_{yr} = C_f M_f a \left(\delta_f - \beta - \frac{a\omega_z}{v_x}\right) \\ + C_r M_r b \left(\beta - \frac{a\omega_z}{v_x}\right). \end{aligned} \tag{12}$$

By integrating (10)–(12), we obtain

$$\begin{aligned} mba_y + I_z\dot{\omega}_z &= m(a+b)a_y \frac{C_f M_f}{\Xi} \\ &\quad + mg\frac{aC_f M_f - bC_r M_r}{\Xi}\sin\phi \\ &\quad + (a+b)\left[\delta_f - \frac{\omega_z(a+b)}{v_x}\right]\frac{C_f M_f C_r M_r}{\Xi}. \end{aligned} \tag{13}$$

In order to facilitate the modeling process of the cornering stiffness and the measured output, we make a reasonable assumption that $C_r = k_c C_f$ and $k_c \in \mathbb{R}^+$ is a ratio constant [17]. (13) can be converted into

$$\begin{aligned} ma_y \frac{k_c b M_r - a M_f}{M_f + k_c M_r} + I_z\dot{\omega}_z - mg\frac{aM_f - bM_r}{M_f + k_c M_r}\sin\phi \\ = \frac{k_c M_f M_r (a+b)}{M_f + k_c M_r}\left[\delta_f - \frac{\omega_z(a+b)}{v_x}\right] \cdot C_f. \end{aligned} \tag{14}$$

When the output and input data are acquired at sample instant k, the cornering stiffness in (13) can be formulated by RLS algorithm

$$Y_m(k) = G^T(k)\Phi(k) + \mathrm{e}(k) \tag{15}$$

where $\Phi(k)$ is the identified parameter of C_f and it is represented as $\hat{C}_f$. $G^T(k)$ is the input regression. $Y_m(k)$ denotes the measured output and expressed as $Y_m(k) = ma_y\frac{kbM_r - aM_f}{M_f + kM_r} + I_z\dot{\omega}_z - mg\frac{aM_f - bM_r}{M_f + kM_r}\sin\phi$.

When the time window length is N, the cost function can be deduced as follows:

$$J_r = \sum_{k=1}^{N} \lambda_m^{N-i}\left[Y_m(k) - G^T(k)\hat{\Phi}(k)\right]^2 \tag{16}$$

where λ_m is the forgetting factor. $\hat{\Phi}(k)$ is estimated as follows:

$$\begin{aligned} \hat{\Phi}(k) &= \hat{\Phi}(k-1) + \mathrm{K}(k)\left[Y_m(k) - G^T(k)\hat{\Phi}(k-1)\right] \\ K(k) &= P(k-1)G(k)\left[\lambda_m I + G^T(k)P(k-1)G(k)\right]^{-1} \\ P(k) &= {\lambda_m}^{-1}\left[P(k-1) - K(k)G^T(k)P(k-1)\right] \end{aligned} \tag{17}$$

where I is the identity matrix, and $K(k)$ and $P(k)$ are the gain matrix and the covariance matrix. We can see that the estimation of the cornering stiffness by RLS does not require the information of the vehicle sideslip and lateral forces.

Furthermore, the rotation model of the wheels is added to describe the vehicle motion, which is specifically presented as

$$I_\omega \dot{\omega} = R_e F_x + T_d - T_b - R_e F_{rr} \tag{18}$$

where I_ω is the wheel inertial moment, ω denotes the wheel angular velocity, R_e is the radius of wheel, F_x is the longitudinal force. T_d and T_b represent the traction torques and brake torques, respectively. F_{rr} is the rolling resistance and it is sufficiently small to be neglected. The tire longitudinal slip ratio is defined as $\sigma = (R_e\omega - v_\omega)/v_\omega$, where v_ω is the velocity on the rolling direction.

C. Kinematic Model and Measurement Equations

The installed inertial measurement unit (IMU) consists of accelerometers and gyroscopes along three different axes, based on which the accelerations and the angle rates of the vehicle can be measured by the configuration. The vehicle movement in the longitudinal direction is obtained in the form of kinematic model

$$\dot{v}_x = a_x^m + \dot{\psi}v_y + g\sin\phi - b_x - r_x \tag{19}$$

where a_x^m stands for the measured longitudinal acceleration, ψ is the yaw angle and equals to the angle between the vehicle's x-axis and the north direction. b_x and r_x represent the bias and uncorrelated noise of sensors. The gravitational acceleration term $g\sin\phi$ is related to the vehicle roll angle. The bias of the inertial sensors can be compensated by the error modeling with LSTM neural networks, then the GPS measurement is exploited to calculate the system states. Specifically, instead of using the wheel speeds, the more accurate horizontal velocity of GPS is adopted since it is independent of the wheel slip. The course angle γ is as follows:

$$\gamma = \arctan\frac{v_{eg}}{v_{ng}} \tag{20}$$

where v_{eg} and v_{ng} are the east and north velocity of GPS in the navigation frame. The GPS ground velocity is $V_{\mathrm{GPS}} = \sqrt{v_{eg}^2 + v_{ng}^2}$. Denote the measurement noise of the course angle as v_G, then it can also be represented as follows:

$$\gamma = \beta + \psi + v_G = \arctan(v_y/v_x) + \psi + v_G \tag{21}$$

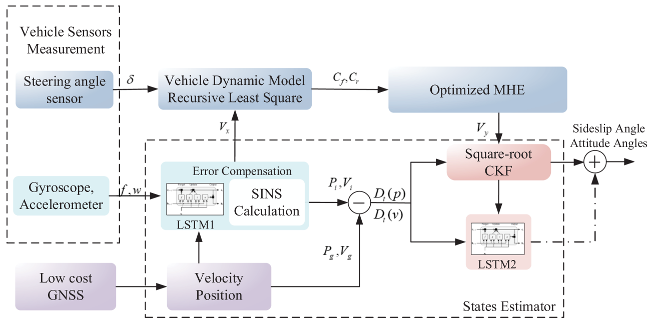

Fig. 2. Structure of the proposed states estimation approach.

where β is the sideslip angle and can be estimated from the three degrees of freedom vehicle dynamic model.

Besides, the equations that correspond to the longitudinal and lateral accelerations are expressed as follows:

$$a_x = \frac{1}{m}(F_{xf}\cos\delta_f - F_{yf}\sin\delta_f + F_{xr})$$
$$a_y = \frac{1}{m}(F_{xf}\sin\delta_f + F_{yf}\cos\delta_f + F_{yr} - mg\sin\phi) \quad (22)$$

On the basis of the above models, a composite nonlinear system model can be derived after combining different parts together. Define the composite state vector $\boldsymbol{x}$ as follows:

$$\boldsymbol{x} = \begin{bmatrix} v_x & v_y & \omega_z \end{bmatrix}^T \quad (23)$$

and based on the analysis for the system, choosing the acceleration as the output variable, while the measured accelerations and steering angle as the input variables, then the input vector $\boldsymbol{u}$ and the observation vector $\boldsymbol{y}$ are defined as follows:

$$\boldsymbol{u} = \begin{bmatrix} \delta & a_x^m & a_y^m \end{bmatrix}^T \quad (24a)$$
$$\boldsymbol{y} = \begin{bmatrix} a_x & a_y & \omega_z \end{bmatrix}^T. \quad (24b)$$

Selecting the sampling time as T_s, then after some mathematical manipulation, the discrete-time system can be finally obtained by the forward Euler method as follows:

$$\boldsymbol{x}_{k+1} = f(\boldsymbol{x}_k, \boldsymbol{u}_k) + \boldsymbol{w}_k$$
$$\boldsymbol{y}_k = h(\boldsymbol{x}_k, \boldsymbol{u}_k) + \boldsymbol{v}_k \quad (25)$$

where $\boldsymbol{x}_k \in \Re^n$, $\boldsymbol{u}_k \in \Re^m$, $\boldsymbol{w}_k \in \Re^n$ is the disturbance vector affecting the system dynamics. $\boldsymbol{y}_k \in \Re^s$, and $\boldsymbol{v}_k \in \Re^s$ is the measurement noise vector. The estimation results include the sideslip angle, which is closely related with different road surface and tire–road interaction coefficient.

III. State Estimator Design

In this section, a nonlinear approach is proposed to achieve accurate estimation for the vehicle states. The structure of the technique is illustrated in Fig. 2, which consists of four different modules as follows:

1) vehicle sensors and global navigation satellite system (GNSS) measurement;
2) an error compensation by information fusion;
3) an optimized MHE;
4) a vehicle sates estimator.

From the figure, we can see that the specific force f and the angular velocity ω_z are obtained from the output of gyroscopes and accelerometers. Then, the bias in inertial sensors can be addressed by the error compensation stage based on LSTM1. The input of the designed network consists of the longitudinal velocity from INS, as well as the lateral velocity error between GPS and IMU of the previous instant. The position and velocity computed from SINS can be expressed as P_i and V_i. Similarly, P_g and V_g are the position and velocity obtained from the GNSS block.

Besides, the lateral velocity is computed from the network output, and the coupling effect between lateral and longitudinal motions can be revealed. Furthermore, the LSTM2 is utilized to optimize the parameters of square-root CKF, $D_t(p)$, and $D_t(v)$ stand for the differences between the position and velocity calculated from INS and GNSS. Meanwhile, the SCKF gains also serve as the inputs of LSTM2 to obtain a final estimation. Finally, we can obtain the estimation of the sideslip angle and attitude angles (roll, pitch, and yaw angle).

The detail of SCKF-based method is described in Algorithm 1. The measurement vector of the extended SCKF is $\boldsymbol{z}$ and expressed as $\boldsymbol{z} = \begin{bmatrix} \boldsymbol{y}^T & \beta & D(p) & D(v) \end{bmatrix}^T$. The LSTM2 is intended to learn and predict the internal filter parameters.

A. Moving Horizon Estimator (MHE)

MHE has been widely used to cope with nonlinear estimation problems with constraints. MHE is composed of a deterministic and a stochastic one. In the first part, the weighting factor depends on the reliability of the measurements. To facilitate description, for the time window of N sampling periods, denote the measurements as $\boldsymbol{y}_{k-N}^k = [y_{k-N}, \ldots, y_k]^T$, the control input as $\boldsymbol{u}_{k-N}^{k-1} = [u_{k-N}, \ldots, u_{k-1}]^T$, and the estimation of $\boldsymbol{x}_{k-N}$ at current instant k as $\hat{\boldsymbol{x}}_{k-N|k}$. Based on the analysis of the system, the cost function J_k is selected as

$$J_k = \rho \left\| \boldsymbol{x}_{k-N|k} - \bar{\boldsymbol{x}}_{k-N} \right\|^2_{P^{-1}_{k-N}} + \sum_{i=k-N}^{k-1} \|\boldsymbol{w}_i\|^2_{Q^{-1}} + \sum_{i=k-N}^{k} \|\boldsymbol{y}_i - h(\boldsymbol{x}_i, \boldsymbol{u}_i)\|^2_{R^{-1}} \quad (26)$$

where $\boldsymbol{x}_k$ and $\boldsymbol{w}_i$ represent the state variable and the process noise, respectively, Q and R stand for the covariances of the process noise and the measurement noise, respectively. The symbol $\|\cdot\|$ represents the Euclidean norm of the vector in $\Re^n$. The scalar $\rho \in [0, 1]$, namely the weight for the arrival cost term, is introduced to enhance the flexibility of the algorithm in practical applications. The expression

$$\bar{\boldsymbol{x}}_{k-N} = \hat{\boldsymbol{x}}_{k-N|k-1} = f(\hat{\boldsymbol{x}}_{k-N-1|k-1}, \boldsymbol{u}_{k-N-1}) + \hat{\boldsymbol{w}}_{k-N-1|k-1} \quad (27)$$

is called a prior state estimate, and the error covariance of $\boldsymbol{x}_{k-N} - \bar{\boldsymbol{x}}_{k-N}$ at time $k-N$ is represented as $P(k-N)$.

The well-known MH estimator is formulated as the optimization problem with a given set $\{\bar{\boldsymbol{x}}_{k-N}, \boldsymbol{y}_k^{N+1}, \boldsymbol{u}_k^N\}$, then $\{\hat{\boldsymbol{x}}_{k-N|k}, \hat{\boldsymbol{w}}_{k-N|k}^{k-1}\} = \mathrm{argminJ_k}$. The variables are subject to the constraint

$$\hat{\boldsymbol{x}}_{i+1|k} = f(\hat{\boldsymbol{x}}_{i|k}, \boldsymbol{u}_i) + \boldsymbol{w}_{i|k}, i = k-N, \ldots, k-1. \quad (28)$$

It can be seen that the cost function reflects the estimation errors defined in a sliding time horizon. In the process, only N+1 measurements are considered to update the arrival cost term. A truncated horizon is selected at the initialization of MHE, then the number of measurements increase with an increasing horizon length when $t \leq N$. Therefore, the MHE is less affected by the inaccurate prior guess when compared with other methods.

B. Optimization Procedure

It has been stated in the former research that the accuracy of the MHE does not enhance apparently with the increase of the horizon length [8]. Nevertheless, the computational cost associates with the horizon window length, more effort should be devoted to reconciling these two objectives. Therefore, some optimization techniques have been developed to improve the computational efficiency and applied to handle the equality constraints in the equations. In order to deal with the larger nonlinear programming brought by the direct multiple shooting, some tools like IPOPT can be exploited in the implementation [38].

In addition to EKF and UKF-based algorithms, some filtering strategies are employed to update the arrival cost in (26). The MHE procedure can be summarized as optimization, propagation and prediction. The optimized Newton moving horizon estimator (OMHE) with multiple iterations is discussed and used to estimate the vehicle states.

In some cases, the model parameters can be optimized by a one-hidden layer feedforward neural network. The inputs are formed by the prediction $\bar{\boldsymbol{x}}_{k-N}$ and the information vector $I_t = col(\boldsymbol{y}_{k-N}, \ldots, \boldsymbol{y}_k, \boldsymbol{u}_{k-N}, \ldots, \boldsymbol{u}_{k-1})$. The output is the neural network estimate $\tilde{\boldsymbol{x}}_{k-N|k}$. When the estimate is obtained from the propagation by the descent steps [39], the function $p_X(x)$ is utilized to calculate the projection on a convex compact set X. Let $\boldsymbol{x}_k \in X \subset \Re^n$, $\boldsymbol{u}_k \in U \subset \Re^s$, and $H : X \times U^{n+1} \rightarrow \Re^q$ be defined as follows:

$$H\left(\boldsymbol{x}_{k-N}, \boldsymbol{u}_{k-N}^k\right) := \begin{pmatrix} h^{\boldsymbol{u}_k} \circ f^{\boldsymbol{u}_{k-1}} \circ f^{\boldsymbol{u}_{k-2}} \circ \cdots \circ f^{\boldsymbol{u}_{k-N}} \\ h^{\boldsymbol{u}_{k-1}} \circ f^{\boldsymbol{u}_{k-2}} \circ \cdots \circ f^{\boldsymbol{u}_{k-N}} \\ \vdots \\ h^{\boldsymbol{u}_{k-N-1}} \circ f^{\boldsymbol{u}_{k-N}} \\ h^{\boldsymbol{u}_{k-N}} \end{pmatrix} \quad (29)$$

where q equals to $(N+1)m$ and $\circ$ denotes function composition. $f^{\boldsymbol{u}_{k-N}}(\boldsymbol{x}_i) \triangleq f(\boldsymbol{x}_i, \boldsymbol{u}_i) + \boldsymbol{w}_i$. Hence, the cost function (26) can be rewritten as follows:

$$J_k = \rho \left\|\boldsymbol{x}_{k-N|k} - \bar{\boldsymbol{x}}_{k-N}\right\|^2_{P^{-1}_{k-N}} + \sum_{i=k-N}^{k-1} \|\boldsymbol{w}_i\|^2_{Q^{-1}} + \left\|\boldsymbol{y}_{-N}^k - H(\boldsymbol{x}_{x_{k-N|k}}, \boldsymbol{u}_{k-N}^k)\right\|^2_{R^{-1}}. \quad (30)$$

Algorithm 1: Extended Square-root cubature Kalman filter.

Input $\hat{\boldsymbol{x}}_{k-1|k-1}$, $\boldsymbol{S}_{k-1|k-1}$, $\boldsymbol{z}_k$;
Initialize sigma-points error matrix;
for $i = 1, \ldots, N$ **do**
Compute the cubature points:
$\boldsymbol{\chi}_{i,k-1|k-1} = \boldsymbol{S}_{k-1|k-1}\xi_i + \hat{\boldsymbol{x}}_{k-1|k-1}$;
The points propagated by system function:
$\boldsymbol{\chi^*}_{i,k|k-1} = f(\boldsymbol{\chi}_{i,k-1|k-1}) + \boldsymbol{w}_{i,k|k-1}$;
State prediction:
$\hat{\boldsymbol{x}}_{k|k-1} = \frac{1}{m}\sum_{i=1}^m \boldsymbol{\chi^*}_{i,k|k-1}$;
State error covariance prediction: $P_{k|k-1} =$
$\frac{1}{m}\sum_{i=1}^m \boldsymbol{\chi^*}_{i,k|k-1}\boldsymbol{\chi^{*T}}_{i,k|k-1} - \hat{\boldsymbol{x}}_{k|k-1}\hat{\boldsymbol{x}}^T_{k|k-1} + Q_{k-1}$
The weighted, centered matrix:
$\boldsymbol{\chi^*}_{k|k-1} = \frac{1}{\sqrt{m}}[\boldsymbol{\chi^*}_{1,k|k-1} - \hat{\boldsymbol{x}}_{k|k-1}\boldsymbol{\chi^*}_{2,k|k-1} - \hat{\boldsymbol{x}}_{k|k-1}\ldots\boldsymbol{\chi^*}_{m,k|k-1} - \hat{\boldsymbol{x}}_{k|k-1}]$;
The square-root of Q_{k-1} is $\boldsymbol{S}_{Q,k-1}$;
$\boldsymbol{S}_{k|k-1} = Tria([\boldsymbol{\chi^*}_{k|k-1}\boldsymbol{S}_{Q,k-1}])$;
Measurement Update:
$\boldsymbol{\chi}_{i,k|k-1} = \boldsymbol{S}_{k|k-1}\xi_i + \hat{\boldsymbol{x}}_{k|k-1}\boldsymbol{Z}_{i,k|k-1} = h(\boldsymbol{\chi}_{i,k|k-1})$
Measurement prediction:
$\hat{\boldsymbol{z}}_{k|k-1} = \frac{1}{m}\sum_{i=1}^m \boldsymbol{Z}_{i,k|k-1}$;
The weighted, centered matrix:
$\boldsymbol{\chi}_{k|k-1} = \frac{1}{\sqrt{m}}[\boldsymbol{\chi}_{1,k|k-1} - \hat{\boldsymbol{x}}_{k|k-1}\boldsymbol{\chi}_{2,k|k-1} - \hat{\boldsymbol{x}}_{k|k-1}\ldots\boldsymbol{\chi}_{m,k|k-1} - \hat{\boldsymbol{x}}_{k|k-1}]$;
$\boldsymbol{Z}_{k|k-1} = \frac{1}{\sqrt{m}}[\boldsymbol{Z}_{1,k|k-1} - \hat{\boldsymbol{z}}_{k|k-1}\boldsymbol{Z}_{2,k|k-1} - \hat{\boldsymbol{z}}_{k|k-1}\ldots\boldsymbol{Z}_{m,k|k-1} - \hat{\boldsymbol{z}}_{k|k-1}]$;
The square-root of R_k is $\boldsymbol{S}_{R,k}$;
$\boldsymbol{S}_{zz,k|k-1} = Tria([\boldsymbol{Z}_{k|k-1}\boldsymbol{S}_{R,k}])$;
The cross-covariance matrix:
$P_{xz,k|k-1} = \boldsymbol{\chi}_{k|k-1}\boldsymbol{Z^T}_{k|k-1}$;
The Kalman gain:
$W_k = (P_{xz,k|k-1}/S^T_{zz,k|k-1})/S_{zz,k|k-1}$;
Updated state:
$\hat{\boldsymbol{x}}_{k|k} = \hat{\boldsymbol{x}}_{k|k-1} + W_k(\boldsymbol{z}_k - \hat{\boldsymbol{z}}_{k|k-1})$;
The square-root of the corresponding error covariance
$\boldsymbol{S}_{k|k} = Tria([\boldsymbol{\chi}_{k|k-1} - W_k\boldsymbol{Z}_{k|k-1}W_k\boldsymbol{S}_{R,k}])$;
end for
return $\hat{\boldsymbol{x}}_{k|k}$;

When the gradient $\nabla J_k(\tilde{\boldsymbol{x}}_{k-N|k})$ is obtained from the above discussion, the computation of cost function at time $k = N, N+1, \ldots$ is deduced as follows:

$$\hat{\boldsymbol{x}}_{k-N|k} = \bar{\boldsymbol{x}}_{k-N|k} - (\nabla^2 J_k(\tilde{\boldsymbol{x}}_{k-N|k}))^{-1}\nabla J_k(\tilde{\boldsymbol{x}}_{k-N|k}). \quad (31)$$

C. Observability and Stability Analysis

It has been proved that the observability of a time-variant system can be derived from the observability Gramian. Therefore, the nonlinear system (25) is locally observable at the operating point $(\boldsymbol{x}_0, \boldsymbol{u}_0)$ if the observability matrix Γ has the same rank as the dimension of state vector n. Γ is defined as follows:

$$\Gamma = \begin{bmatrix} \frac{\partial L_f^0 h}{\partial x} & \frac{\partial L_f^1 h}{\partial x} & \ldots & \frac{\partial L_f^{n-1} h}{\partial x} \end{bmatrix}^T_{(\boldsymbol{x}_0, \boldsymbol{u}_0)}$$

$$:= \begin{bmatrix} J_{L_f^0 h} & J_{L_f^1 h} & \cdots & J_{L_f^{n-1} h} \end{bmatrix}^T_{(\boldsymbol{x}_0, \boldsymbol{u}_0)} \tag{32}$$

where $L_f^i h$ denotes the ith order Lie derivative, and $J_{L_f^i h}$ is the corresponding Jacobian matrix.

When (1)–(25) are combined, the first two elements in Γ are computed as follows:

$$J_{L_f^0 h} = \begin{bmatrix} J_{0,11} & J_{0,12} & J_{0,13} \\ J_{0,21} & J_{0,22} & J_{0,23} \\ 0 & 0 & 1 \end{bmatrix} \tag{33}$$

$$J_{L_f^1 h} = \begin{bmatrix} J_{1,11} & J_{1,12} & J_{1,13} \\ J_{1,21} & J_{1,22} & J_{1,23} \\ J_{1,31} & J_{1,32} & J_{1,33} \end{bmatrix} \tag{34}$$

where $J_{0,ij}$ and $J_{1,ij}$ represent the elements of the Jacobian matrices. It can be inferred from (32) that the maximum rank of $J_{L_f^0 h}$ is 3. Since the state vector dimension in (25) is 3, the system is validated to be locally observable.

In order to analyze the stability of the estimation methods, some assumptions are briefly introduced. Especially, when the functions $f(\cdot)$ and $h(\cdot)$ in (25) are assumed as class C^2, and $f(\cdot)$ satisfies the Lipschitz condition, we have

$$|f^u(x') - f^u(x'')| \leq k_f \, |x' - x''| \, \forall x,' x'' \in X. \tag{35}$$

Then, the Hessian matrix of the cost function (30) is represented as $\nabla^2 J_k(\hat{x})$. From the above discussion, it can be concluded that there exists $k_J > 0$ such that the Lipschitz inequality (36) holds for all time instant.

$$\left|\nabla^2 J(x') - \nabla^2 J(x'')\right| \leq k_J \, |x' - x''| \, \forall x,' x'' \in X. \tag{36}$$

Based on the theory of iterative Newton, the update for the MHE estimate at time $k+1$ is as follows:

$$\hat{x}_{k-N+1|k+1} = \bar{x}_{k-N+1|k+1} - (\nabla^2 J_{k+1}(\tilde{x}_{k-N+1|k+1}))^{-1} \times \nabla J_{k+1}(\tilde{x}_{k-N+1|k+1}). \tag{37}$$

Then, the estimation error e_{k-N+1} is obtained and (38) can be established by using a Taylor expansion centered in x_{k-N+1} and related assumptions

$$\begin{aligned} |e_{k-N+1}| &\leq \frac{(k_f k_J \, |e_{k-N}| + 2\rho) \, k_f}{2\rho} \, |e_{k-N}| \\ &= \alpha_{k-N+1} \, |e_{k-N}| \leq {\alpha_0}^{k-N+2} \, |x_0 - \bar{x}_0| \, . \end{aligned} \tag{38}$$

Therefore, the estimation error for vehicle states given by (31) is stable if $\alpha_0 = \frac{(k_f k_J |x_0 - \bar{x}_0| + 2\rho) k_f}{2\rho} < 1$.

D. LSTM Neural Networks

The LSTM neural network is introduced to solve the gradient problems in traditional RNNs. In the designed neural network, a memory cell is added in the hidden layer and four elements are included: An input gate (i_t), a neuron with a self-recurrent connection, a forget gate (f_t), and an output gate (o_t). The (39)–(44) are derived to describe how the information flow is updated at every time step

$$f_t = \mathrm{sigmoid}(W_f h_{t-1} + U_f x_t + b_f) \tag{39}$$

$$i_t = \mathrm{sigmoid}(W_i h_{t-1} + U_i x_t + b_i) \tag{40}$$

$$\tilde{c}_t = \tanh(W_c h_{t-1} + U_c x_t + b_c) \tag{41}$$

$$c_t = f_t \circ c_{t-1} + i_t \circ \tilde{c}_t \tag{42}$$

$$o_t = \mathrm{sigmoid}(W_o h_{t-1} + U_o x_t + b_o) \tag{43}$$

$$h_t = o_t \circ \tanh(c_t) \tag{44}$$

where x_t is the input at time step t, b_f, b_c, b_i, b_o are the bias vectors. W_f, W_c, W_i, W_o are the weight matrices. $\circ$ means element-wise multiplication and sigmoid$(\cdot)$ denotes the element-wise sigmoid function. In the first step, the input gate i_t and the estimated states $\tilde{c}_t$ of memory cells at time t are obtained by (40) and (41). In the second step, f_t is computed by sigmoid function and then, the state c_t is updated by integrating $i_t, \tilde{c}_t, f_t$ in (39). At last, the output gate value h_t is derived with the updated states.

From the illustration of the proposed approach in Fig. 2, the LSTM1 is designed to model the INS error and LSTM2 can optimize the parameters of ESCKF. The detailed input and output of the networks are described in (42). In LSTM2, the differences between the position and velocity calculated from INS and GNSS are represented as $D_t(p)$ and $D_t(v)$. Meanwhile, the SCKF gains also serve as the inputs of LSTM2 to obtain a final estimation. Therefore, we can obtain the estimation of the sideslip angle and attitudes.

It has been proved that the LSTM can be used to learn and predict the filter parameters, so that the results achieved by ESCKF is robust for different scenarios. The number of layers and nodes in LSTM1 and LSTM2 are the same. The number of hidden layers is 20 and the initial learning rate is 0.1. A decay is set at 0.03 after the second patch, and the weight penalty is 0.01. The initial number of neurons is 150 and the threshold of the target error is 0.02.

IV. Simulation and Experimental Results

In this section, the effectiveness of OMHE and the proposed methods described in the previous sections are validated by Carsim-MATLAB simulation and field tests. In the simulation, different driving manoeuvres are designed and the vehicle state measurements generated from the trajectories are taken as the system input. Besides, the results obtained from Kalman-based, MHE, and OMHE methods are analyzed accordingly. In the field test, a comprehensive comparison of the estimation of the vehicle sideslip angle and states are preformed.

A. Simulation Results

Two different simulation scenarios are designed, including a modified double lane change (DLC) and a sine wave steering condition. In the Carsim software, a hatch-back vehicle model is employed and the vehicle parameters are set as follows: $m = 1270$ kg, $I_z = 1343$ kg $\cdot$ m^2, $a = 1.02$ m, and $b = 1.56$ m. In the modified DLC simulation, the steering angle changes with the vehicle trajectory and the speed is 30 km/h. The sideslip angle and estimation errors computed by Kalman, MHE, and OMHE are given in Fig. 3. It can be seen that OMHE achieves higher accuracy than that of Kalman and MHE, especially when

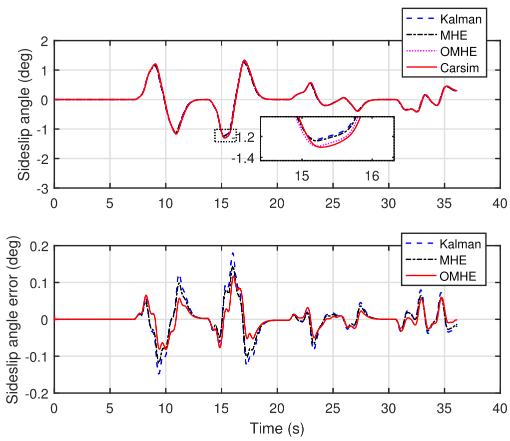

Fig. 3. Sideslip angle and estimation errors in DLC simulation.

TABLE I
VEHICLE SIDESLIP ANGLE ESTIMATIONS BY DIFFERENT METHODS

Scenario (Speed, Maneuvers)	RMSE of estimation error (Units: °)		
	Kalman	MHE	OMHE
30 km/h, modified DLC	0.0448	0.0356	0.0286
0-30 km/h, sine wave	0.0171	0.0130	0.0083

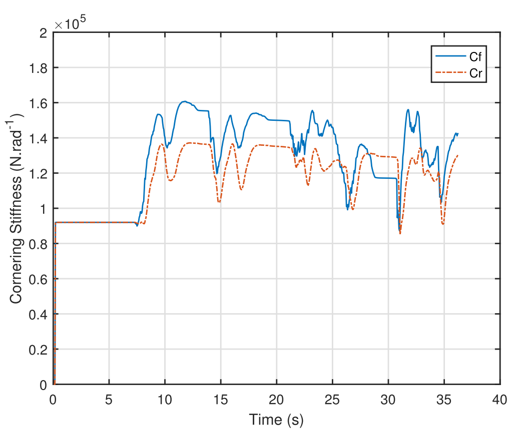

Fig. 4. Cornering stiffness estimation based on RLS in DLC simulation.

the steering angle changes at 10 s and 17 s. The root mean square errors (RMSE) of the results are listed in Table I, which suggest that RMSE of MHE is decreased by 20%, compared with that of OMHE. In the sine wave simulation, the steering angle varies with the vehicle trajectory and the range of speed is 0–30 km/h. The friction coefficient is assumed as a constant and set as 0.85 in the simulation test.

Similarly, the sideslip angle and estimation errors obtained by Kalman, MHE, and OMHE are shown in Fig. 5. The RMSE of the different approaches are summarized in Table I; it can be seen that the error of MHE is reduced by 36.15%. In addition, the cornering stiffness parameters during the simulation are obtained by RLS and depicted in Figs. 4 and 6. It is worth noting

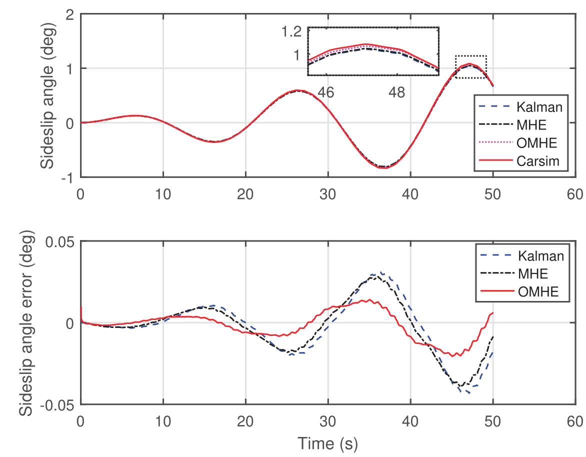

Fig. 5. Sideslip angle estimation in sine wave simulation.

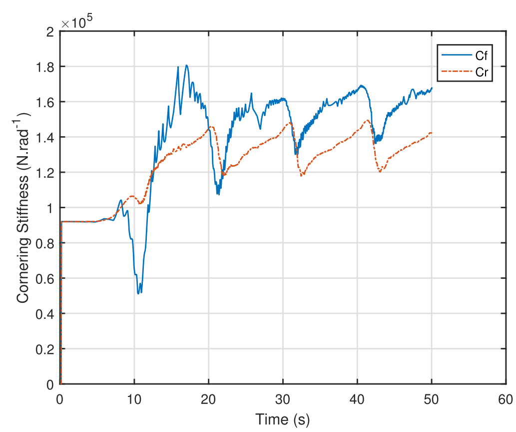

Fig. 6. Cornering stiffness estimation based on RLS in sine wave simulation.

that the variations in the estimated front and rear cornering stiffness are affected by the sideslip angle, but they are almost similar during different driving conditions. The proportional assumption about the parameters are reasonable and can be used to facilitate the modeling process. Furthermore, the stability and accuracy of the OMHE approach is enhanced by introducing the iteration decent algorithm, whereas the performance of KF and MHE rely heavily on parameters.

B. Field Tests and Discussion

In the field test, a class sedan installed with a differential GPS is utilized to measure the sideslip angle. The reference value can be computed as follows:

$$\beta = \frac{2+(-1)^j}{2}\pi + (-1)^i\varphi - \gamma \tag{45}$$

where i denotes the order of quadrant number. $j = 1$ when $i = 1, 2$, and $j = 2$ when $i = 3, 4$. γ is the angle between the longitudinal speed and the north direction. φ is the vehicle heading direction.

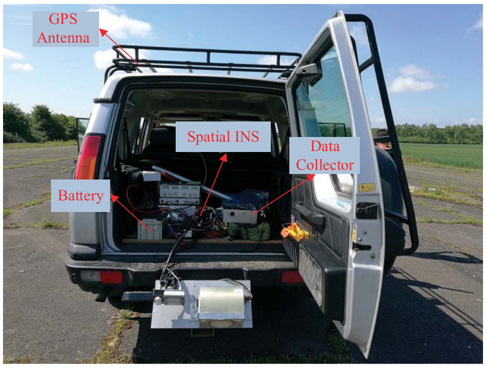

Fig. 7. Experimental setup of integrated navigation system.

TABLE II
SPECIFICATIONS FOR INTEGRATED NAVIGATION SYSTEM

INS parameters	Value
IMU output rate	200 Hz
Gyroscope bias instability	0.05 $°/h$
Gyroscope Scale factor	$\leq$50 ppm
Accelerometer bias instability	15 μg
GPS output rate	20 Hz

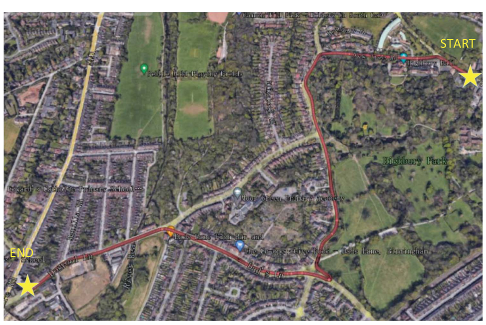

Fig. 8. Test vehicle trajectory.

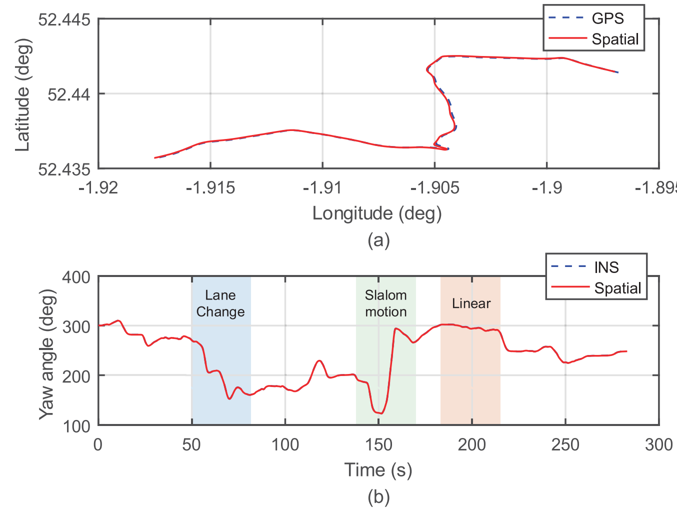

Fig. 9. Longitude, latitude and yaw angle of the vehicle in the field test.

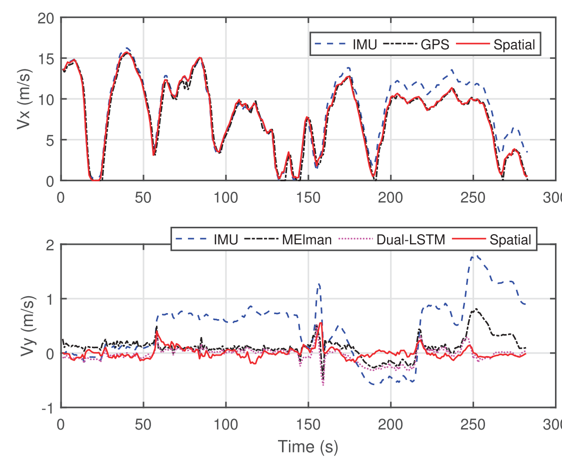

Fig. 10. Longitudinal and lateral velocity estimated by IMU, GPS, modified ENN, LSTM, and spatial FOG.

To verify the performance of the proposed method in estimating the sideslip angle and vehicle states under real driving conditions, a GPS/IMU integrated navigation system is installed on the vehicle and the experimental setup is shown in Fig. 7. The advanced navigation Spatial fiber optic gyroscope (FOG) system consists of KVH 1750 IMU and Trimble BD920 GPS with sampling rates of 200 Hz and 20 Hz, respectively. The specifications about the integrated navigation system are described in Table II. A synchronization strategy is employed to deal with the different sampling rates problem [40]. The vehicle trajectory is designed as Fig. 8 and lasts for 283 s.

It can be seen that several straight lines, DLC and similar slalom maneuvers under different conditions are included during the movement. The longitude and latitude of the vehicle recorded by spatial FOG and GPS are given in Fig. 9(a). The yaw angle is depicted in Fig. 9(b) and changes dramatically at about 60 s and 150 s, which can be considered as the lane change (blue areas) and slalom maneuvers (green areas). Besides, the vehicle linear motion is related to the approximate constant yaw angle. The stability of the estimation method can be verified under different scenarios in real applications.

The longitudinal and lateral velocities estimated by different methods are illustrated in Fig. 10. From the longitudinal velocity figure, we can see that the highest speed exceeds 50 km/h, and the velocity changes more dramatically than that of the simulation. In addition, the comparison of different lateral velocities by the inertial sensors, the modified ENN (MElman), and the proposed dual-LSTM method are also clearly illustrated in the figure. The results show that the proposed dual-LSTM method has a better ability to model the relationship of the longitudinal velocities computed by IMU and the low cost GPS.

In the network structure design, the longitudinal velocity error and lateral velocity error at previous instant are taken as the input of the neural network, then the prediction at current instant can be obtained from the network output. The initial learning rate is set as 0.1 with a decay of 0.02 from the fifth epoch. The number

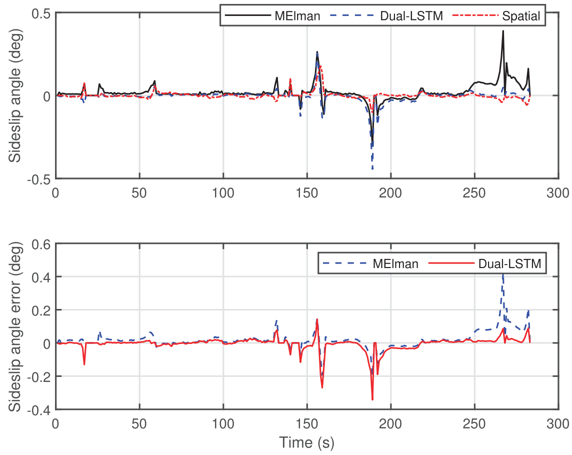

Fig. 11. Sideslip angle estimated by IMU, GPS, Modified ENN, LSTM, and spatial FOG.

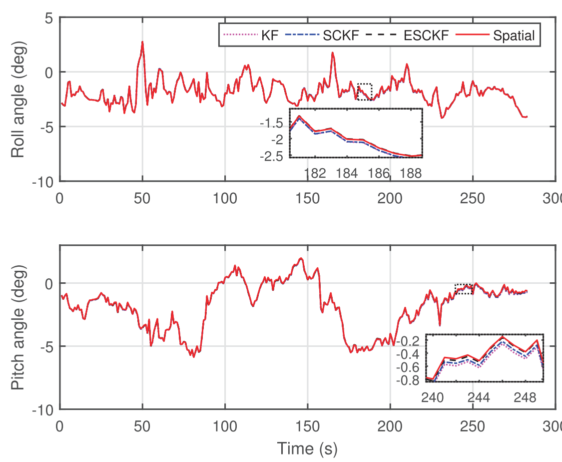

Fig. 12. Roll and pitch angles estimated by KF, SCKF, ESCKF, and Spatial.

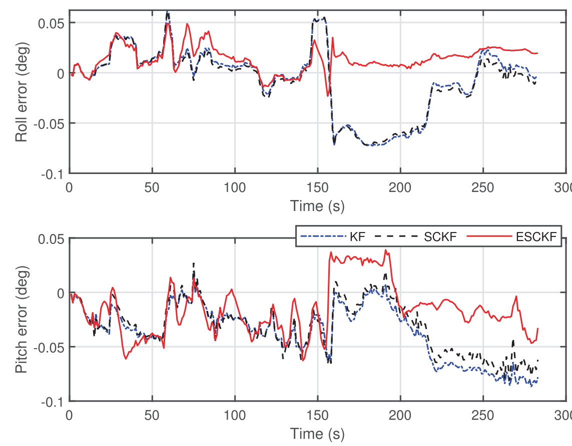

Fig. 13. Roll and pitch angle errors estimated by KF, SCKF, and ESCKF.

TABLE III
COMPARISON OF VEHICLE STATES ESTIMATION ERROR

Vehicle states	RMSE of estimation errors (Units: °)			
	INS	Kalman	SCKF	ESCKF
Roll angle	0.038	0.034	0.032	0.017
Pitch angle	0.062	0.044	0.039	0.029
Yaw angle	0.165	0.164	0.164	0.158

of hidden layer is 50 and the maximum iterations are set as 600. Therefore, the estimation accuracy of lateral velocity and sideslip angle of the vehicle can be improved by eliminating the measurement errors. Finally, the sideslip angles are obtained by different methods and the estimation errors of MElman and dual-LSTM are depicted in Fig. 11. It can be concluded that the dual-LSTM has better ability to estimate the sideslip angle under different driving scenarios. Optimal performance is obtained by the designed structure since the memory cell is incorporated to determine the input information and the INS errors are compensated by LSTM1, which thus improves the estimation accuracy when compared with other methods.

The estimation outputs from spatial are taken as the reference values to validate the results obtained by different methods. The roll and pitch angles measured by Spatial during the experimental test are depicted in Fig. 12. Furthermore, the roll and pitch angle estimation errors by INS, KF, SCKF, and extended SCKF (ESCKF) are plotted in Fig. 13, we can see that the angle errors of ESCKF are remarkably reduced from 160 s. The vehicle longitudinal velocity drops rapidly to 0 at about 190 s, then rises quickly and stabilizes around 40 km/h from 200 to 250 s, as shown in Fig. 10. Therefore, it implies that the developed method is more robust and reliable to the velocity variation. The reason is that the filter parameters are optimized by the LSTM structure and the weights in the self-loop can be changed dynamically. This allows the method to learn and predict the internal filter parameters and enhances its ability to estimate the attitude angles under different scenarios.

The RMSE of estimation errors are listed in Table III. It can be summarized that RMSE of the roll angle and pitch angle by ESCKF decreases by 46.7% and 25.6%, respectively. Therefore, the experimental results indicate that the proposed dual-LSTM and ESCKF are effective in improving estimation accuracy of the vehicle sideslip angle and states.

V. CONCLUSION

This article provides a novel framework based on OMHE, optimized neural networks, and ESCKF to estimate the automotive sideslip angle and states. In particular, the INS is used to measure the lateral and longitudinal information, and their coupling effects are revealed accordingly by introducing a LSTM with powerful modeling and prediction abilities. Furthermore, the OMHE is designed to deal with the time-varying steering angle under various driving scenarios, and the sideslip angle can be estimated precisely. Besides, the other LSTM is adopted to learn and predict the internal filter parameters of ESCKF, and the vehicle states can be obtained. Finally, the simulation and field tests are implemented to validate the effectiveness of the proposed method. The results show that the designed approach has a better estimation accuracy with respect to the existing methods, especially when executing different maneuvers including a straight driving, a lane change, and so on.

In the future, in order to deal with different road conditions and GPS denied environment [41], [42], we need to investigate the approach to enhance the stability and navigation accuracy. Some adaption strategies and diagnosis modules will be considered to detect the availability of GPS signal and improve the robustness of the system. Besides, the generalization of LSTM should also be studied in some complex driving conditions. Furthermore, the states estimation approach can be extended to the research of autonomous navigation of mobile robots, as well as related areas [43].

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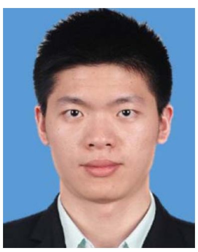

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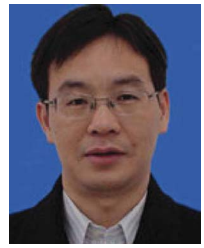

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